

F24-194-D-NewsNet : GraphRAG leveraging KGraphs for deep news analysis using LLMs

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Chapter 1

Introduction

This document outlines the requirements and specifications of a NewsNet, A system that focuses on causal news detection

The purpose of this document is to discuss a detailed report on the GraphRAG (Retrieval-Augmented Generation) NewsNet project. Primarily, the proposal focuses on a concise introduction and description of the project. First, after a discussion on the problem, it elaborates on the proposed solution, highlighting stakeholders and the motivation behind the idea. Furthermore, this document aims to aid the understanding the project description including scope of the project, working modules, tools/technology used, division of labor and tailored timeline.

1.1 Problem Statement

Many business and financial decisions rely on news for crucial predictions and decision-making processes.[1] Learning about a significant event through the news requires one to search far and wide for why the event took place and what led to such an event occurring. Searching for a series of causes and effects, one finds themselves reading article after article from different resources to identify the chain of events. This laborious task of scouring the internet to find relevant text in the era of Artificial Intelligence is counter intuitive. Furthermore, this widens the gap between user interest and the relevancy of the news especially when encountering several various sources of news for the required information.

1.2 Scope

Causal relationship detection through causal arguments by feeding into a knowledge graph to then help an LLM (Large Language Model) leveraging GraphRAG (Retrieval Augmented Generation) to retrieve a Cause/Effect sequence of events chained together giving a new sense of depth to news.

Scraping unstructured news text data from the web to extract meaningful causal relationships. Adhering by an ontology of news to find specific repeating relationships to create a graph. Applying a model approach on extracting relationships. Detecting events and assigning them as anchor points in our knowledge graph to make sense of the news and how different entities have a causal relationship among them. Querying the said knowledge graph to find various relationships in depth depending on type of query. Thus aiding the LLM to know where to look and how entities have a deeper relationship with one another than how it seems on the surface. Searching based on category and applying user chosen geographic filters to further create user tailored recommendations and results.

1.3 Modules

1.3.1 Data Collection and NEER

Web Scraping and Data Annotation for data ingestion to train a Named Entity and Event Recognition model.

1. Internet Scraping and labelling data.
2. Named Entity and Event Recognition Model.

1.3.2 Causal Knowledge Graph

Constructing Knowledge Graph using Causal relationships among entities found in the data.

1. Causal relationship identifier.
2. Causal Knowledge Graph Creation.

1.3.3 GraphRAG

Implementation of GraphRAG

1. Fine-tuning LLM according to news data.
2. Using causal KGraph to enhance retrieval.

1.3.4 GraphRAG vs Baseline RAG

Testing capabilities of the GraphRAG.

1. GraphRAG vs Baseline RAG
2. Testing against various metrics.

1.3.5 Chatbot Creation

Creating a scaffolding structure for the Large Language Model in the Web Application.

1. Creating a chat bot Layout for our LLM.
2. Smaller Knowledge graph creation for visual summary of user query.
3. Timeline Layout Creation.
4. Article Reference.
5. Trend Analysis.

1.3.6 Quality Assurance

Web Application Quality Assurance

1. Performance Metrics.
2. Unit Testing.
3. System Testing.
4. Scraping latest information for Model updation.

1.3.7 User Experience Optimization

Data Authentication, Category Based Search and User tailored recommendations

1. Outlier Flagging and Data point distance calculation.
2. Category based searching.
3. User tailored recommendations.

1.4 User Classes and Characteristics

User Classes and Characteristics

Table 1.1: User Classes and Characteristics

User class	Description
News Viewer	A News Viewer is an individual who regularly engages with news outlets to consume news content and stay upto date on current events related to their topic of interest. 15 - 20 % of the population reads newspapers online in Pakistan. We expect about a a 1000 people to interact with our application to access it once a day to get an update for current events. The interaction is mostly on website to read current news and headlines.
History Learner	A History Learner dives into studying past events and their backgrounds. This group has about a few thousand people, including students, teachers, and those who teach themselves. History Learners often use the app several times a week. They look for detailed articles and resources to better grasp stories from the past. They like a simple-to-use interface that makes it easy to find chosen content, timelines, and learning materials.
Researcher	A Researcher works in academia or industry and needs trustworthy detailed information for their work or publications. This group has about 1,500 people, including academics grad students, and experts in various fields. Researchers often turn to the platform to find solid data, papers, and sources. They need powerful search tools and ways to sort through scholarly articles and old records. Many use the platform to manage citations and look at data.
Media Enthusiast	A Media Enthusiast loves media trends, production, and how people communicate. This group has about 800 people, including future journalists, bloggers, and students studying communication. Media Enthusiasts are likely to use the app every week to learn about media analysis, critiques, and what's new in the industry. They enjoy content they can interact with, like podcasts and online seminars, and might join in talks or chat rooms on the platform.

Chapter 2

Project Requirements

2.1 Use-case/Event Response Table/Storyboarding

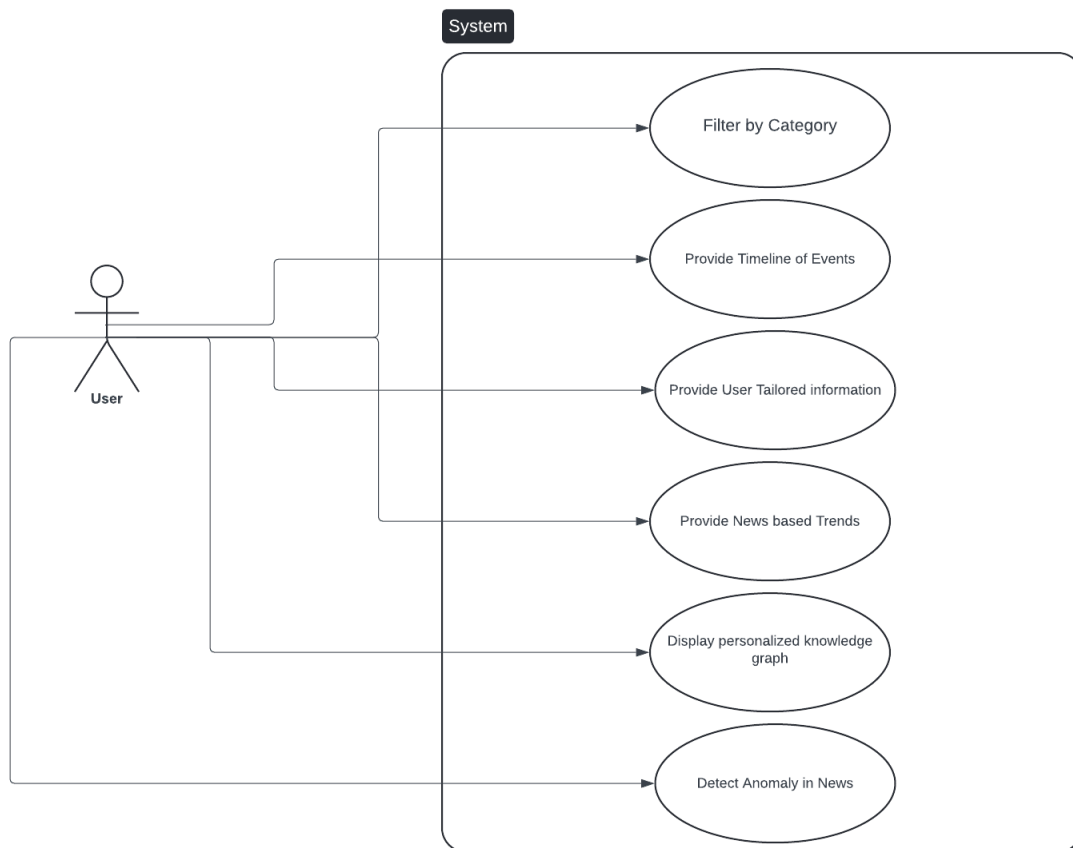


Figure 2.1: Use case Diagram

2.1.1 Filter by category

Use case name	Filter By Category
Scope	System under design
Level	User level
Primary Actor	User
Stakeholders and Interests	User: Wants to access relevant news articles according to a certain category.
Preconditions	The user is on the relevant page in the application.
Success guarantee (postcondition)	The user is shown the relevant news articles that he has requested, filtered, and selected according to the category specified.

Main Success Scenario	
Actor's Actions	System's Responses
1. User selects the options bar.	2. System presents a list of options in a drop-down to select a category.
3. User selects the needed category from the options available.	4. System displays the articles in result according to the user's choice.

Extensions	
Actions	Responses
1. No articles are found for the user's selected category.	1. The user is notified about the lack of results for the category.

2.1.2 Provide Timeline of Events

Use case name	Provide Timeline of Events
Scope	System under design
Level	User level
Primary Actor	User
Stakeholders and Interests	User: Wants to access relevant news articles in a chronological order of occurrence .
Preconditions	The user is on the relevant page in the application.
Success guarantee (postcondition)	The user is shown the relevant news articles that he has requested, arranged in sequence by a timeline.

Main Success Scenario	
Actor's Actions	System's Responses
1. User presses the options button.	
2. User selects the needed option from the choices available.	3. System displays the articles in result according to a temporal sequence.

Extensions	
Actions	Responses
1. No articles are found for the user's selected options.	1. The user is notified about the lack of results for the time period.

2.1.3 Provide User tailored Information

Use case name	Provide User Tailored Information
Scope	NewsNet
Level	User level
Primary Actor	User
Stakeholders and Interests	User: Wants to news articles and recommendations according to their interests.
Preconditions	The user is on the relevant page in the application. The user has a history of using the application.
Success guarantee (postcondition)	The user receives customized recommendations of news articles.

Main Success Scenario	
Actor's Actions	System's Responses
1. User selects the option to view recommendations.	2. System recommends articles based on the categories and type of news searched before by the user.

Extensions	
Actions	Responses
1. No articles are found for the user.	1. The user is notified about the lack of results due to lack of history.

2.1.4 Provide news based on Trends

Use case name	Provide News based Trends
Scope	NewsNet
Level	User level
Primary Actor	User
Stakeholders and Interests	User: Wants to view and visualize trends according to news data.
Preconditions	The user is on the relevant page in the application.
Success guarantee (postcondition)	The user receives trends and patterns according to his preferred news.

Main Success Scenario	
Actor's Actions	System's Responses
1. User presses the options button.	
2. User selects the needed option from the choices available.	3. System displays the information required in text and graphical representation.

Extensions	
Action	Response
1. Insufficient data for the information provision.	1. The user is notified about the lack of results due to lack of historical news data for the category.

2.1.5 Display personalized knowledge graph

Use case name	Display personalized knowledge graph
Scope	NewsNet
Level	User level
Primary Actor	User
Stakeholders and Interests	User: Wants to visualize their searches and results in a graphically rich format.
Preconditions	The user is on the relevant page in the application. The user has a history of using the application.
Success guarantee (postcondition)	The user receives a knowledge graph that shows their searches and topics explored.

Main Success Scenario	
Actor's Actions	System's Responses
1. User selects the option of viewing the knowledge graph	2. System displays the knowledge graph according to the recent searches.

Extensions	
Action	Response
1. Insufficient data for knowledge graph generation.	1. The user is notified about the lack of results due to lack of history of usage.

2.1.6 Detect anomaly in news

Use case name	Detect anomaly in news
Scope	NewsNet
Level	User level
Primary Actor	User
Stakeholders and Interests	User: Wants to be notified regarding any significant anomaly in the news data.
Preconditions	The user is on the relevant page in the application. The user has a history of using the application.
Success guarantee (postcondition)	The user is notified of anomalies in the provided news information.

Main Success Scenario	
Actor's Actions	System's Responses
1. User initiates anomaly detection on the news provided.	2. System displays the anomalies and inconsistencies in the news.

Extensions	
Action	Response
1. No significant or probable anomaly found	1. The user is notified about the authenticity of information.

2.2 Functional Requirements

2.2.1 Data Collection and Named Entity and Event Recognition (NEER)

1. The system shall collect data from the News website through web scraping tools.
2. The system shall pre-process and clean the data.
3. The system shall implement Named Entity and Event Recognition model on the news data.

2.2.2 Causal Knowledge Graph

1. The system shall identify causal relationships among the entities in the data.
2. The system shall construct a causal knowledge graph (KGs) using the causal relationships.
3. The system shall update and add new relationships to update the knowledge graph as more data is available.

2.2.3 GraphRAG

1. The system shall use the data from the news to fine-tune a Large Language Model (LLM).
2. The system shall use the causal knowledge graph to improve and enhance information retrieval using Graph Retrieval-Augmented Generation (GraphRAG).

2.2.4 GraphRAG vs Baseline RAG

1. The system shall implement a baseline Retrieval-Augmented Generation (RAG).
2. The system shall utilize the baseline RAG to perform a performance comparison between RAG and a GraphRAG.
3. The system shall test the system performance of GraphRAG using predefined metrics.

2.2.5 Chatbot Creation

1. The system shall create a chatbot Interface that provide options to users to select news.
2. The system shall create and provide a personalized knowledge graph to the user according to their query.
3. The system shall display the news results according to the relative timeline.
4. The system shall provide references for the articles in which the news was provided.
5. The system shall provide a trend analysis based on the news.

2.2.6 Quality Assurance

1. The system shall be monitored and tested according to performance metrics.
2. The system shall be tested to verify the functionality of each module.
3. The system shall have system-wide testing to ensure all functionality and their integration as well as performance.

2.2.7 User Experience Optimization

1. The system shall provide options for searching using different categories.
2. The system shall recommend news to users according to their search history.

2.3 Non-Functional Requirements

2.3.1 Reliability

REF-1: The system will provide a detailed log of errors in case of system failure.

REF-2: The system shall provide an error window to the user that will provide the nature of the failure.

REF-3: The system shall be designed by a strategy that will minimize the percentage of error or system failure to less than 5%.

2.3.2 Usability

USE-1: The system user interface (UI) will be highly intuitive, allowing users to use the application without external help.

USE-2 : The system will allow the user to search up relevant news with a single interaction.

USE-3 : The system shall provide a simple guide for users to be able to utilize the application with ease.

2.3.3 Performance

PER-1: The system will load different modules in the application within 3 sec.

PER-2: The system shall utilize fast retrieval to and high speed databases to ensure high

performance.

PER-3: The system will utilize top of the line models and libraries to ensure quick processing and minimum delays.

2.3.4 Security

SEC-1: The system will be secure against breach in user data in storage or transit.

SEC-2: The system will respond appropriately to requests that surpass a normal human standard.

2.4 Domain Model

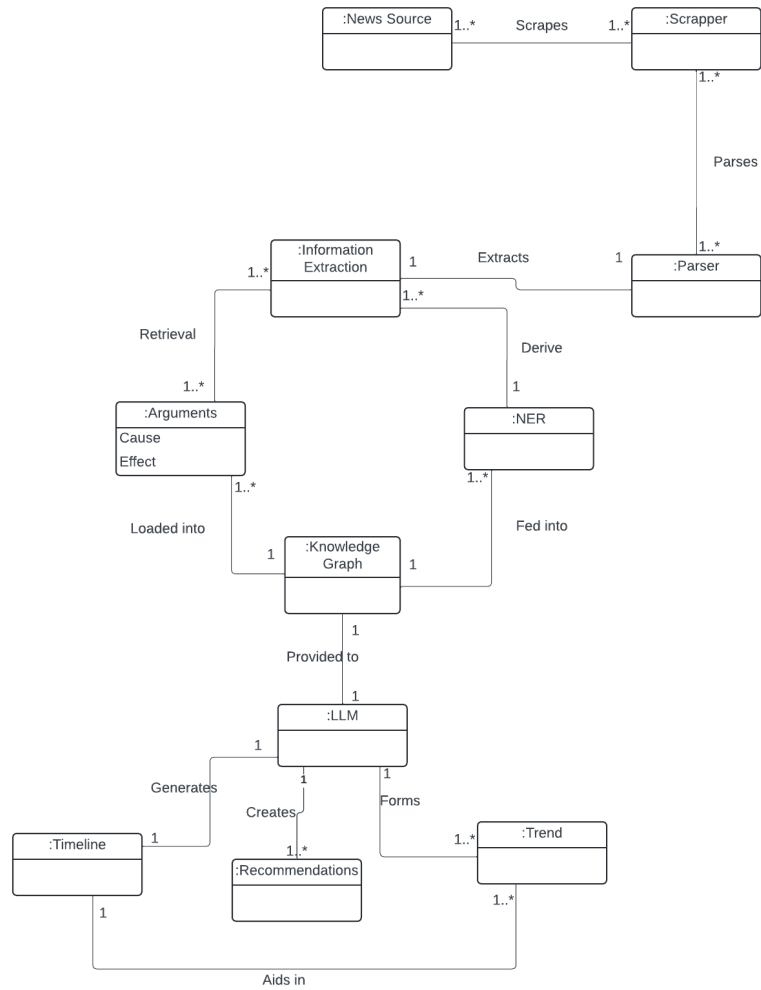


Figure 2.2: Domain Model

Chapter 3

System Overview

NewsNet aims to deliver a system that builds causal relationships in news articles using knowledge graphs. It addresses the complexity of news events and the need for automated tools to analyze vast amount of news data to provide users a better understanding of the cause-effect relationships between events. The system follows a client-server architecture which focuses on entity detection, graph generation and user interaction by providing a user-friendly platform for journalists, researchers and news analysts.

3.1 Architectural Design

The architecture of NewsNet is designed based on Client-Server model. The modular system is composed of multiple interconnected subsystems. The design is structured in order to extract and handle large amount of news data using Natural Language Processing(NLP) techniques to highlight the casual relationship between events.

Major subsystems:

1. Data Collection:

Scraping news articles from different online sources (to ensure continuous data flow)

2. NER and Event Extraction:

Processing collected news data to extract key events and their respective relations using NER (Named Entity-Event Recognition)

3. Casual Graph generation:

Building a casual knowledge graph by identifying relationships between events (based on NER output.)

4. Knowledge Graph(KG) Visualization:

Displaying an interactive interface for users to explore the cause-effect relationships.

5. User Interaction and Search:

Provides search functionality for querying the knowledge graph for improved results.

Box and Line Diagram showing the major subsystems and their connections.

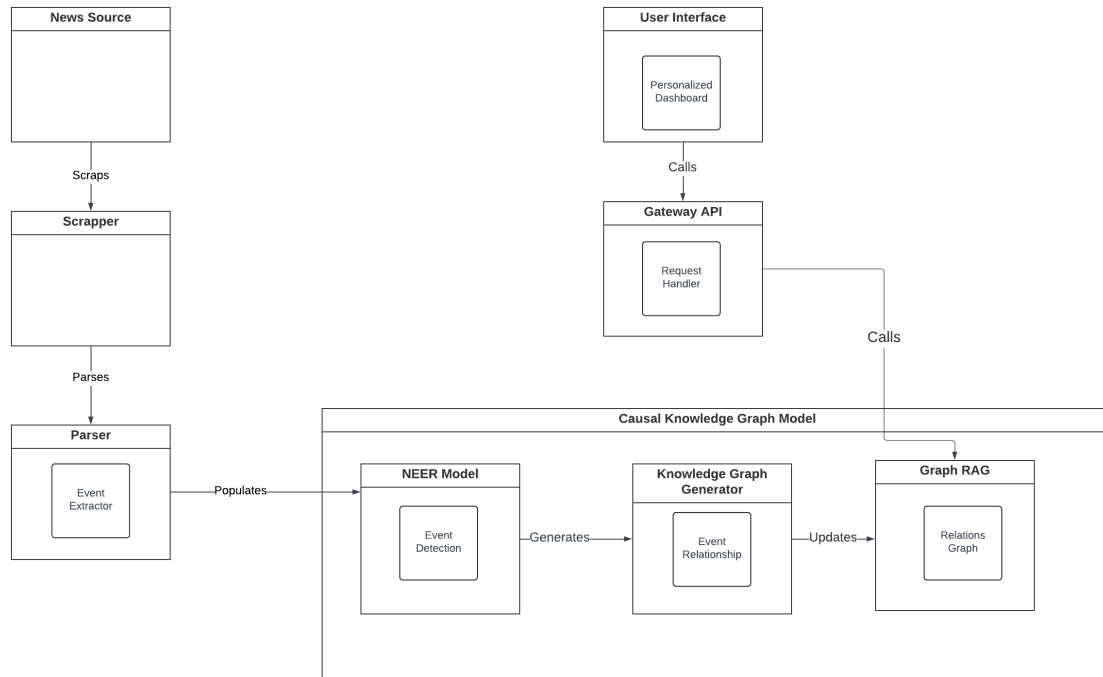


Figure 3.1: Box and Line Diagram

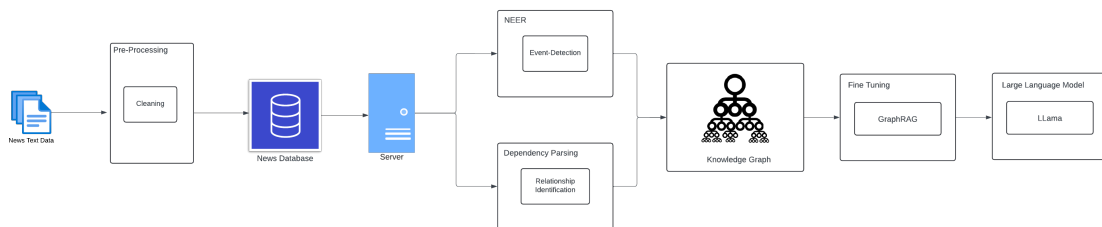


Figure 3.2: Architecture Diagram

3.2 Design Models

Design Models for Object Oriented Development Approach

3.2.1 Activity Diagram

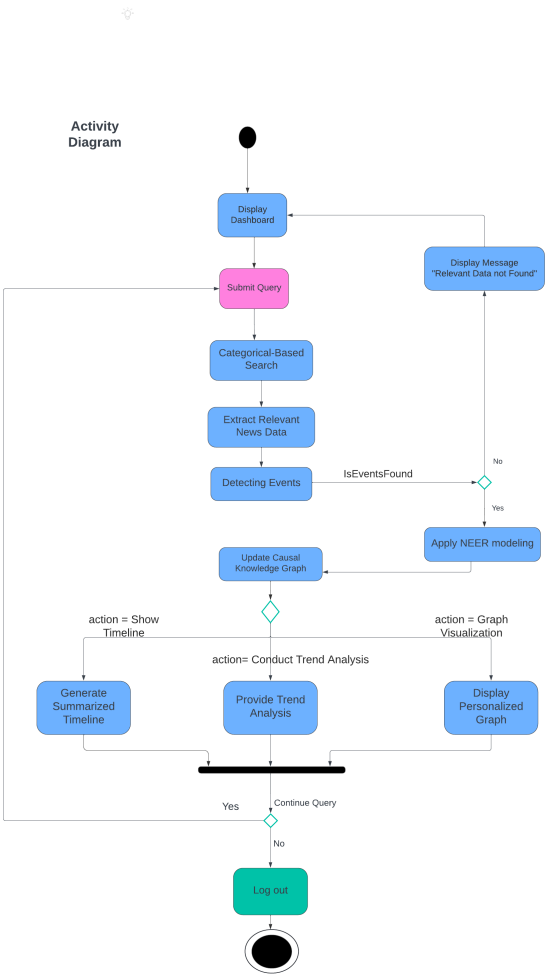


Figure 3.3: Activity Diagram

3.2.2 Class Diagram

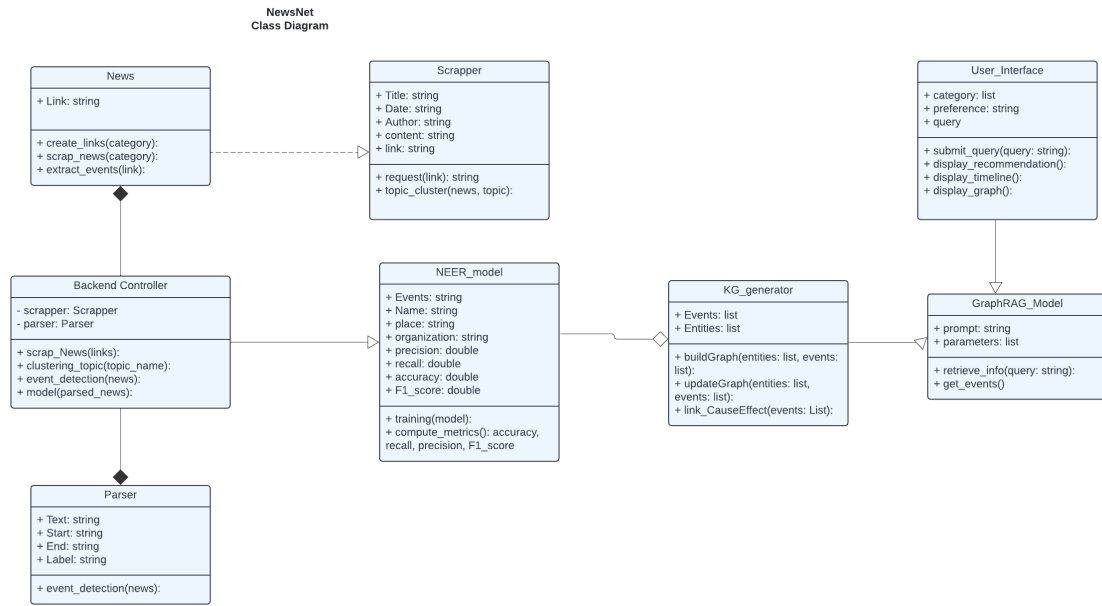


Figure 3.4: Class Diagram

3.2.3 Sequence Diagram

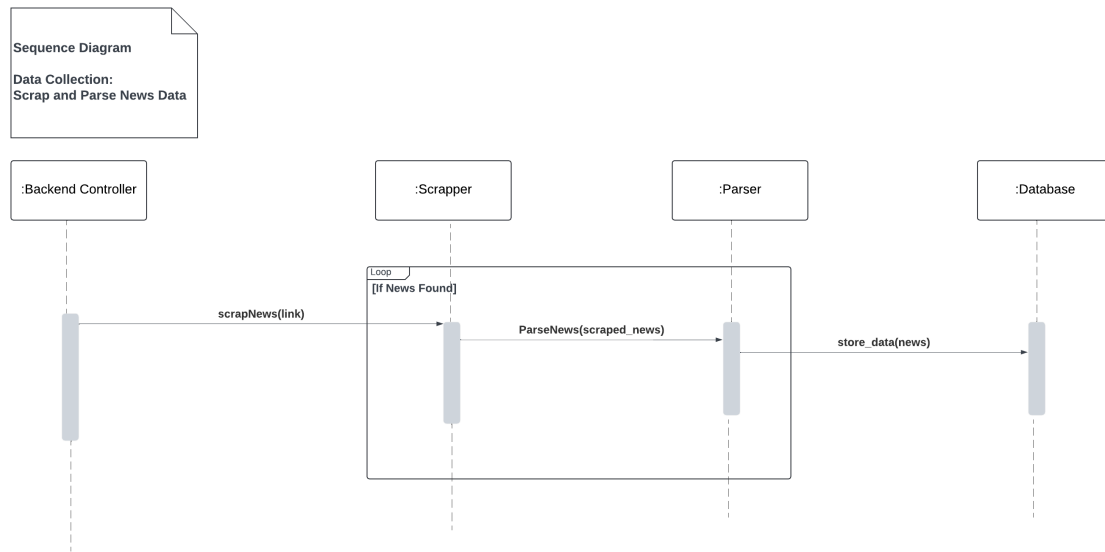


Figure 3.5: Sequence Diagram: 1. Data Collection

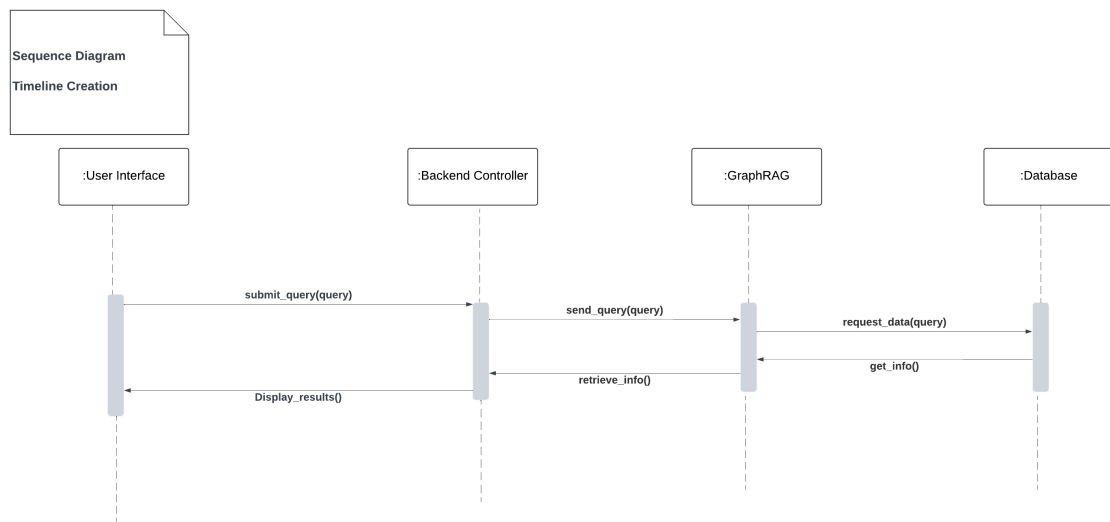


Figure 3.6: Sequence Diagram: 2. Information Retrieval

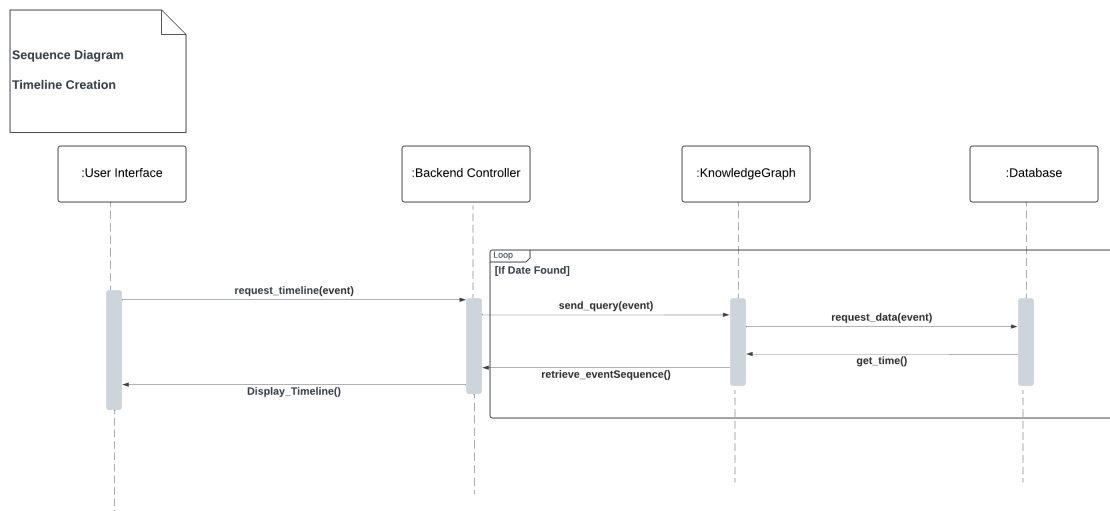


Figure 3.7: Sequence Diagram: 3. Timeline Creation

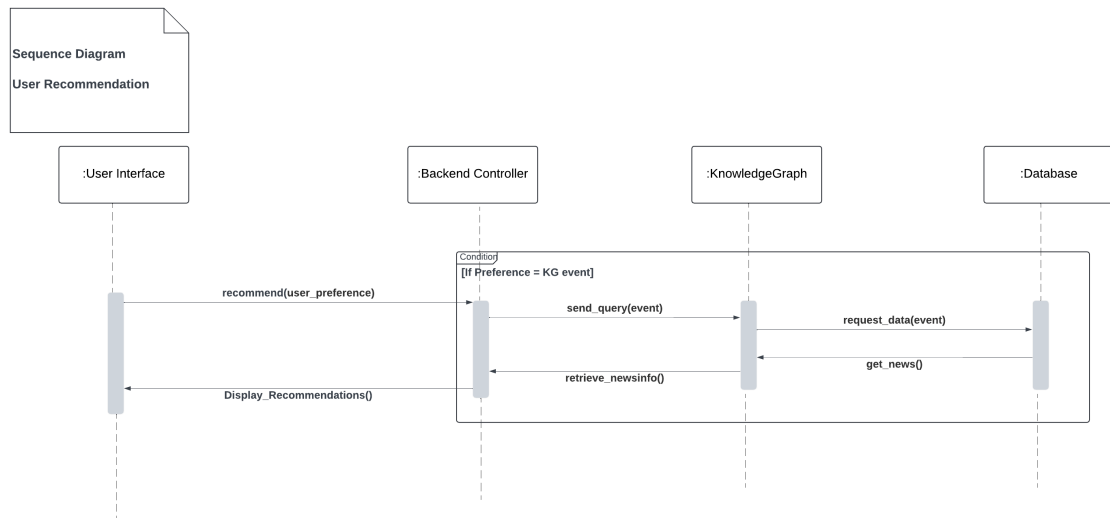


Figure 3.8: Sequence Diagram: 4. User Recommendations

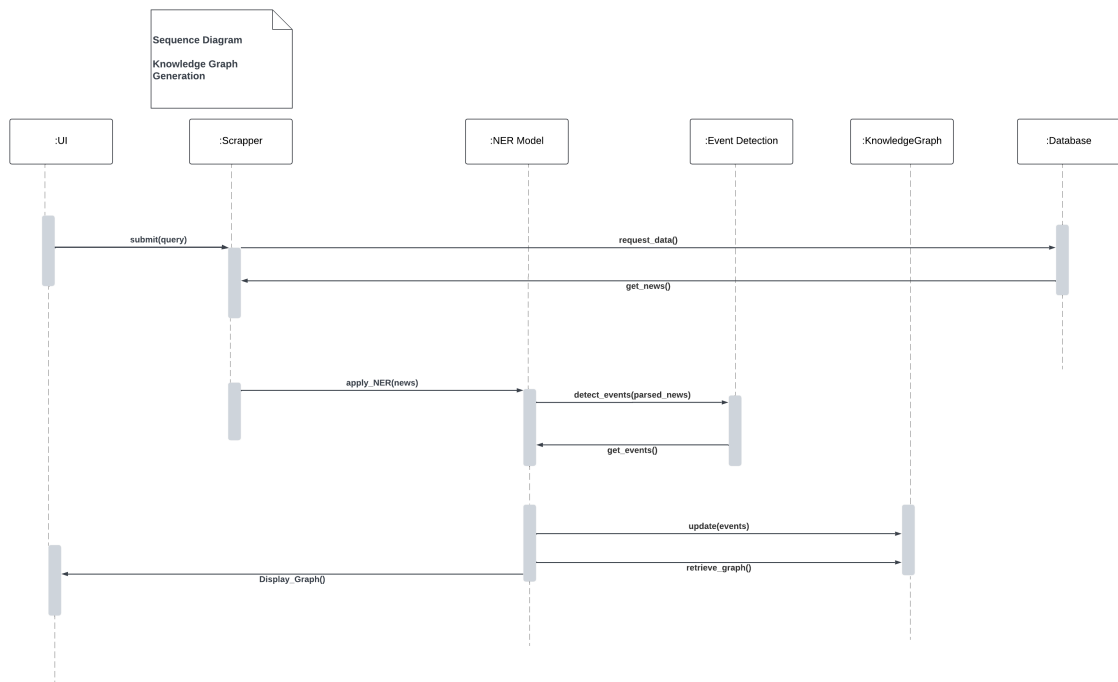


Figure 3.9: Sequence Diagram: 5. Knowledge Graph Generation

3.2.4 State Transition Diagram

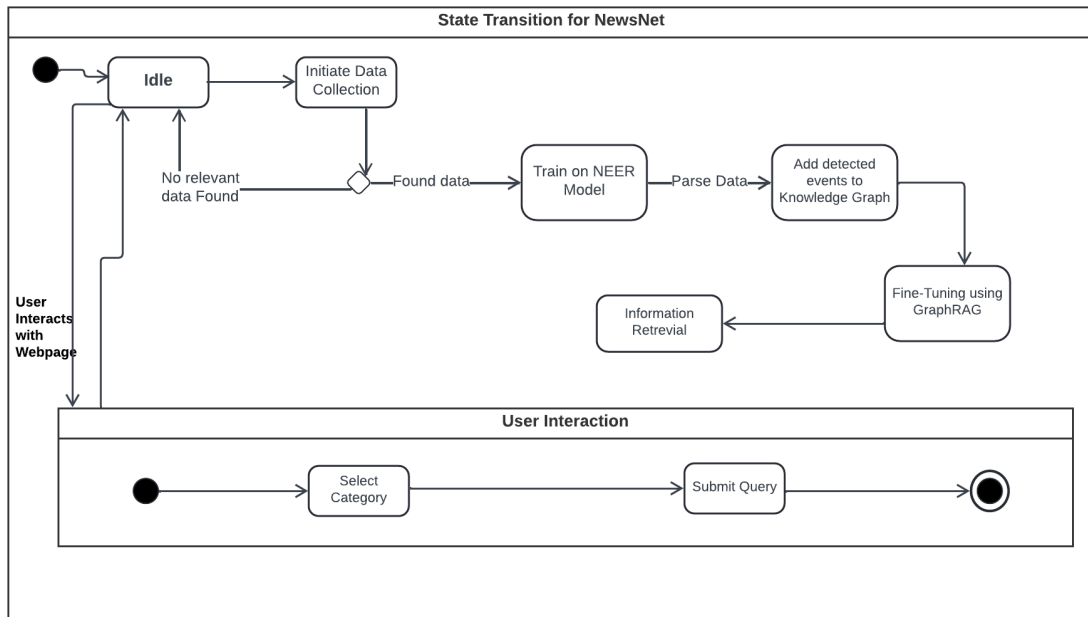


Figure 3.10: State Transition Diagram

3.3 Data Design

The data design of NewsNet primarily focuses on storing and organizing information from the news articles about events and their relationships to make it easy to access and visualize. While json files act as database for storing and managing structured details like article titles, news content and publication dates, a graph database aids in handling connection between different events and entities.

3.3.1 ERD

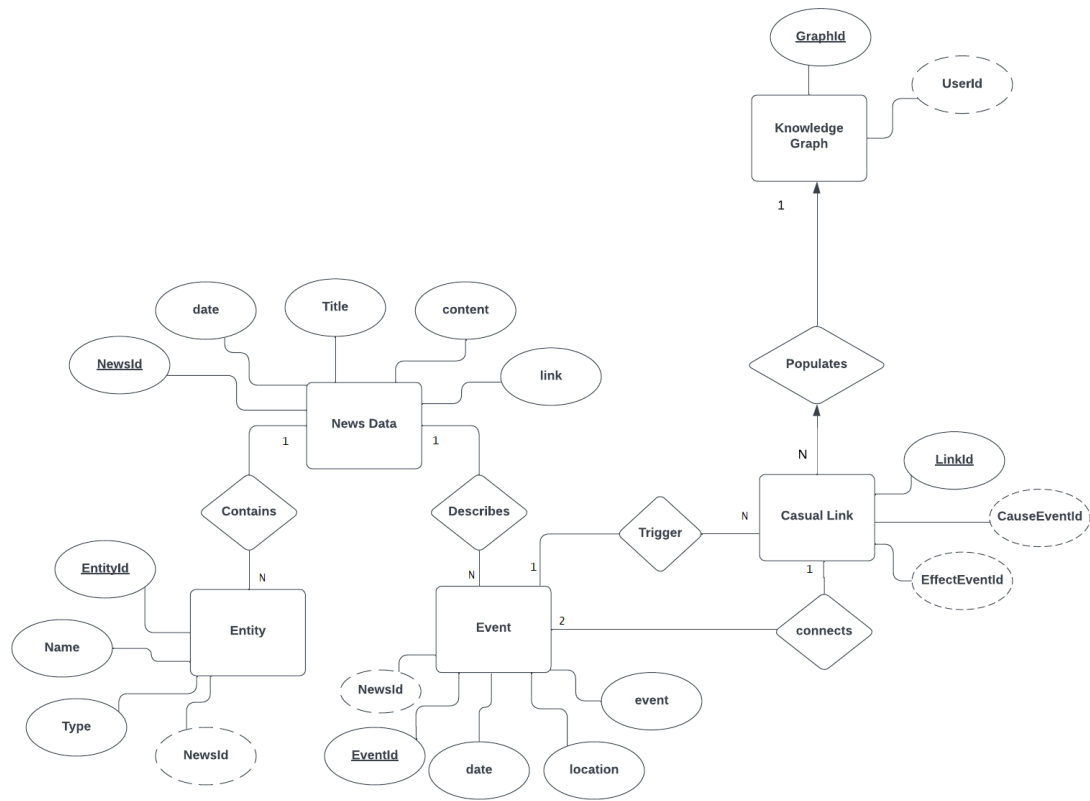


Figure 3.11: Entity Relationship Diagram

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- [1] Ebrahimi Kahrizsangi K. Aghaei Chadegani A. Sharifi, E. The impact of media news on investors' decision- making according to their degree of risk-taking. *Mathematical Modeling and Analysis*, 2024.